

HARSHAD SHINDE

Full Stack Developer | Backend Engineer

Pune, Maharashtra

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PROFILE

Results-driven Python Full Stack Developer with 8+ months of professional experience building enterprise applications using Python, Django, Vue.js, and PostgreSQL. Hands-on experience developing AI-powered applications using LLM APIs, RAG, vector search, prompt engineering, and Groq AI. Experienced in REST API development, database design, retrieval pipelines, and AI application workflows. Published researcher with 3 peer-reviewed papers in AI applications and healthcare technology.

TECHNICAL SKILLS

- Programming Languages: Python, Java, JavaScript, C, C++
- Backend & Web Frameworks: Django, Django REST Framework, FastAPI, Flask, Spring Boot
- Frontend & UI Development: Vue.js, React, Next.js, HTML5, CSS3, Bootstrap.
- Databases & Data Management: PostgreSQL, MySQL, Database Design, Query Optimization, Data Analysis.
- AI/ML & Large Language Models: Retrieval-Augmented Generation (RAG), Vector Search, Hybrid Search, Reranking, Prompt Engineering, Groq AI API, OpenAI API, NLP, Machine Learning.
- Development Tools & Practices: REST APIs, Microservices Architecture, Git/GitHub, Agile/Scrum, System Design, API Documentation, Database Migration.
- Cloud & Deployment: Supabase, Vercel, CI/CD.

CORE COMPETENCIES

RAG Systems • LLM Integration • Hybrid Retrieval • Vector Search • REST API Design • Database Design & Optimization • Prompt Engineering • Backend Architecture • System Design • Performance Optimization • Enterprise Software Development

EMPLOYMENT HISTORY

Python Full Stack Developer

KodSquad Technologies Pvt. Ltd.

Jan 2026 – Aug 2026

- Developed and enhanced enterprise immigration case management modules using Python, Django, and Django REST Framework, implementing complex business requirements and production-ready backend functionality.
- Engineered dynamic questionnaire workflows with conditional branching, hierarchical question structures, ERI/PERI (Entity Relationship Information/Package Entity Relationship Information) mapping.
- Implemented questionnaire mapping and resolution logic to dynamically determine applicable questions, options, forms, and conditions based on case and packet configurations.
- Developed and maintained RESTful APIs with request validation, exception handling, database integration, and seamless Vue.js frontend communication.
- Designed and optimized PostgreSQL database queries, ORM operations, relationships, and data retrieval logic for complex application workflows.
- Debugged and resolved complex production issues involving database relationships, questionnaire rendering, conditional logic, mapping resolution, data consistency, and backend application behavior.
- Collaborated with development teams on feature implementation, debugging, testing, code reviews, and integration of backend and frontend.

Java Full Stack Intern

Archer Infotech

Aug 2024 – Nov 2025

- Developed backend services using Java and Spring Boot, contributing to RESTful API development and backend functionality.
- Implemented and tested REST APIs while gaining hands-on experience with backend integration, debugging, and application development.
- Gained practical experience in Java Full Stack Development, Spring Boot, REST APIs, database management, software development lifecycle, and Git-based development workflows.

EDUCATION

B.E. in Artificial Intelligence and Data Science

2021 – 2025

Dr. D. Y. Patil College of Engineering & Innovation, Talegaon, Pune

- Specialization: Machine Learning | Deep Learning | Data Science | Big Data Analytics | Software Development
- Key Coursework: Artificial Intelligence | Natural Language Processing | Computer Vision | Database Management Systems | Web Application Development | Distributed Computing
- CGPA: 7.71

KEY PROJECTS

Enterprise AI Knowledge & Operations Copilot — In Progress

Python, FastAPI, PostgreSQL, RAG, Vector Search, LLMs, Groq AI, REST APIs

- Developing an enterprise AI knowledge assistant using Python, FastAPI, PostgreSQL, and Retrieval-Augmented Generation (RAG) to provide grounded answers from organizational knowledge.
- Implemented a hybrid retrieval pipeline combining vector and keyword search with configurable candidate retrieval and reranking for improved document relevance.
- Built retrieval evaluation experiments covering Recall@K, Precision@K, MRR, NDCG, latency, reranker candidate depth, and final Top-K selection across 33 evaluation cases.
- Implemented query rewriting, multi-query retrieval, query decomposition, contextual conversation handling, and clarification workflows for complex enterprise questions.
- Added citation generation and source validation, allowing responses to reference the underlying knowledge-base chunks used for generation.
- Developed configurable RAG parameters for candidate retrieval depth and final Top-K results, with automated tests for retrieval pipeline behavior.

AI-Driven Healthcare Diagnosis and Treatment System

Python, Flask, MySQL, HTML, CSS, Bootstrap, REST APIs, Machine Learning Models.

- Developed a Healthcare AI and disease prediction system using Python and Machine Learning, applying Supervised Learning algorithms including Random Forest and Support Vector Machine (SVM) for disease classification.
- Implemented an end-to-end machine learning pipeline covering Data Preprocessing, Feature Engineering, Model Training, Predictive Modeling, and Model Evaluation to improve disease classification performance.
- Integrated personalized healthcare recommendations based on Predictive Modeling results, providing an intelligent decision-support experience for users.
- Built a scalable Flask REST API for symptom input, data processing, predictive analysis, and disease prediction, with API Optimization techniques to improve application responsiveness.

AI Resume Analyzer & Career Optimization Platform

Next.js, React, Supabase, Groq AI, REST APIs, PostgreSQL, Authentication.

- Developed an AI-powered SaaS platform using Python, LLMs, and Groq AI for intelligent resume analysis, ATS optimization, and personalized career coaching.
- Engineered a resume parsing pipeline using NLP and Machine Learning techniques to extract skills, work experience, education, and certifications from uploaded resumes.
- Implemented an ATS Optimization and scoring algorithm to identify critical keyword gaps, evaluate resume relevance, and generate actionable improvement recommendations
- Integrated Groq AI APIs with Prompt Engineering techniques to generate personalized interview questions, career insights, and coaching feedback based on resume content.

RESEARCH EXPERIENCE

- Machine Learning in Healthcare: Developing an AI Assistant Doctor IJSREM Vol. 08, Issue 11, 2024. DOI: 10.55041/IJSREM38474
- AI-Assisted Creativity: Balancing Innovation and Authenticity IJACECT, Vol. 14, Issue 01, 2025.
- Intelligent Medication Recommendation Systems IJACECT, Vol. 14, Issue 01, 2025.
- Copyright Registration — Govt. of India